Where does Qwen3-32B *hold* a paper's authors?

A layer-by-layer readout of author names for 9,108 papers

Stage 10 · jlens

Qwen3-32B · 2026-10-26

The question

target (4,621 papers)
can't write the authors
but picks the true list 4/4 times

control (2,220 papers)
can't write the authors
and can't pick the true list

If the model already **knows** target authors in its middle layers and only fails to say them, a lens on those layers should read the right name **better for target than control**.

If the knowledge is only assembled at the end, the gap should appear only near the output.

So: at which layer does target pull ahead of control?

The data

9,108 papers, 8 fields

33,811 author names

149,999 name tokens scored

13 citation bins (0 to 63,901)

Groups (from an earlier labelling run of the same model)

target 4,621 · control 2,220

guard (writes the authors) 214 · excluded 2,053

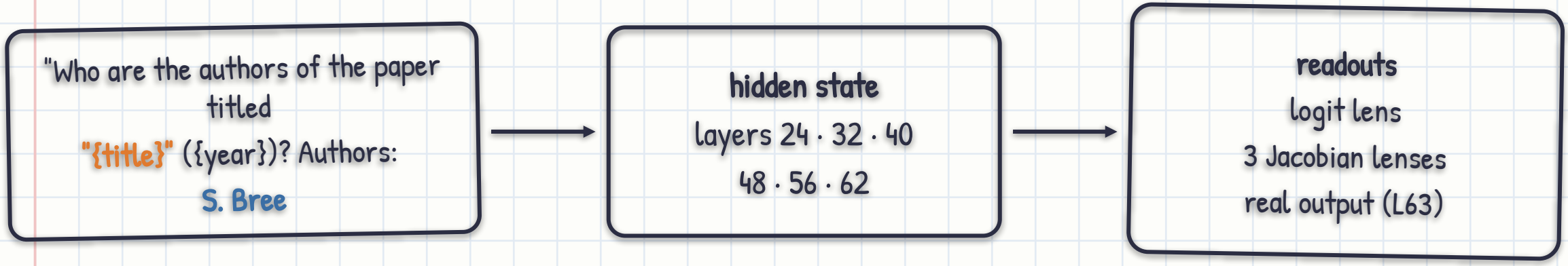
Medicine, biology, CS, psychology, chemistry, maths, physics, materials science: ~1,100 papers each

Name formats

73% start with an initial ("S. Bree")

26% start with a full given name ("Yan Zhao")

How we read the model



Score = rank of the correct next name token,

as log10(1 / rank): 0 = rank 1, -1 = rank 10

7 title conditions: the true title, 5 matched false titles, no title

title-specific signal

= true title - mean(5 false titles)

Same name, same tokens: what is left is knowing

this paper

Target vs control is always compared within a citation bin. 95% bootstrap CIs over papers (2,000 draws).

Can we trust the numbers?

fast = slow

One forward pass vs one per prefix: max gap 0.37 logit (bf16 rounding); ranks 1-41 identical

matches Stage 6

972 tokens: same token ids, median log-prob gap ~1e-5

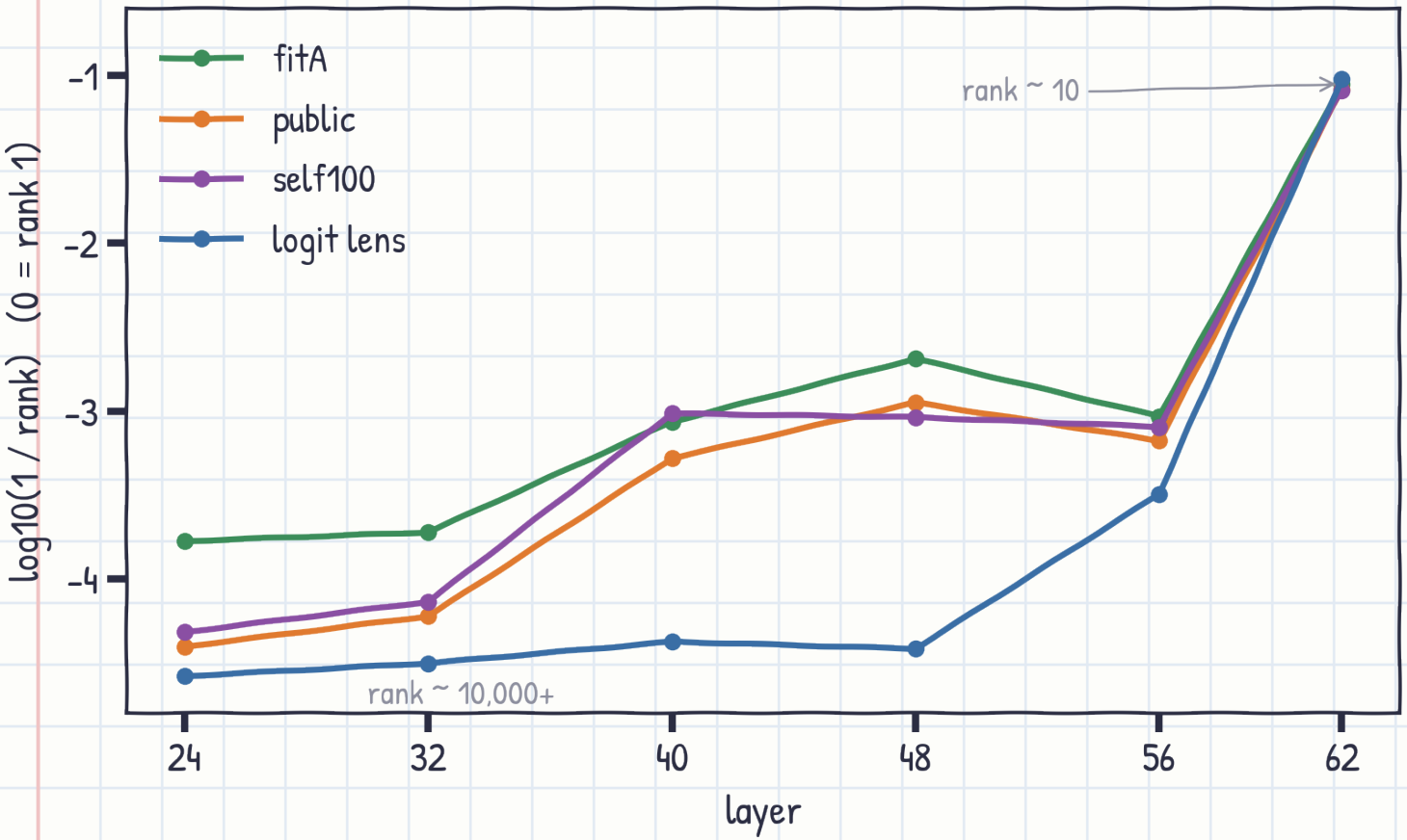
negative control

Punctuation tokens show ~0 title-specific signal at every layer

Caveat: the parity tolerance was raised 0.25 -> 0.5 after seeing these bf16 gaps.

Names become readable late

all 9,108 papers · true title



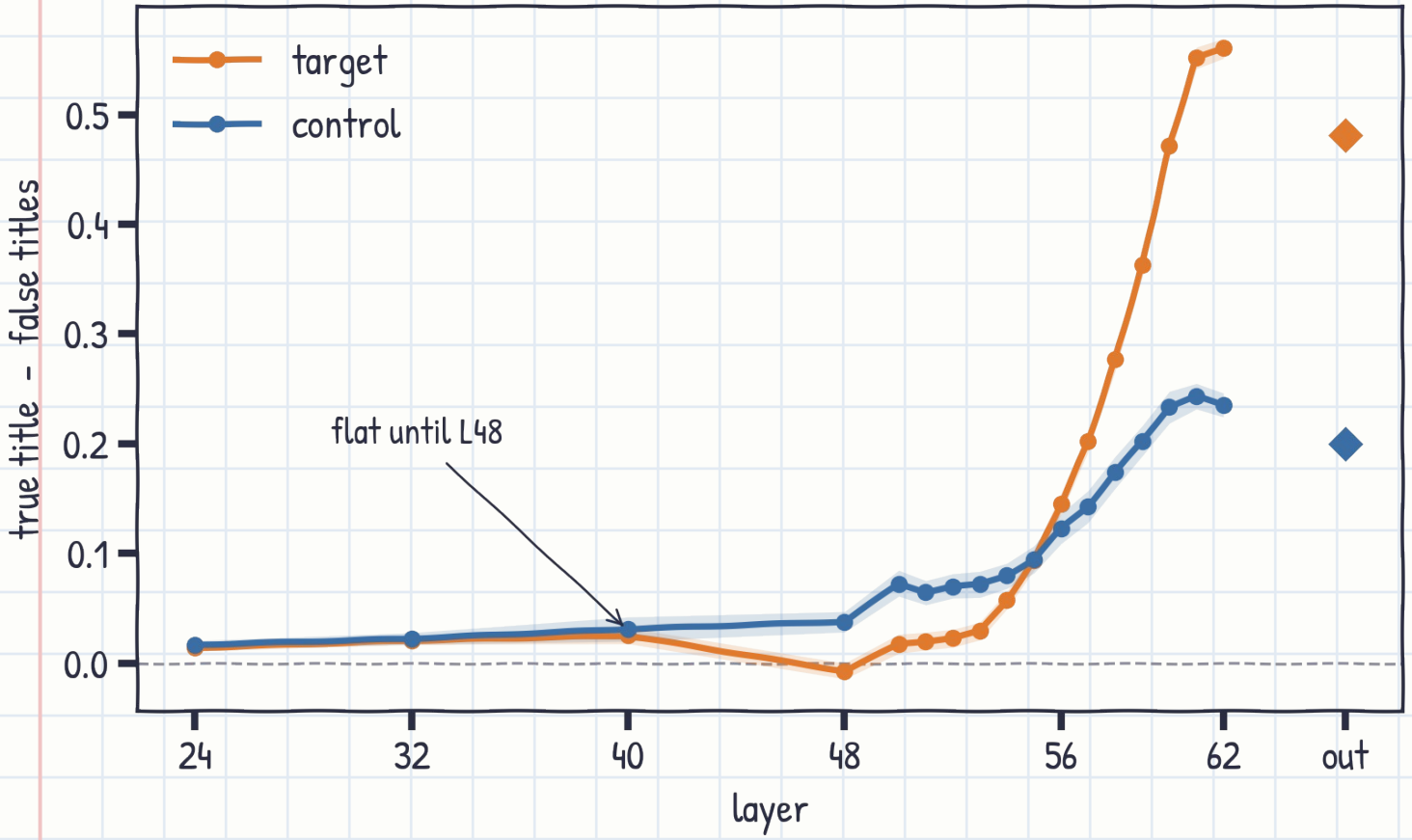
Logit lens: almost nothing until **L48**

Jacobian lenses read more from **L40**

Everything jumps at **L62** (rank ~10)

Knowing the paper shows up only at the end

title-specific signal · public Jacobian lens (lines) and model output (diamonds)



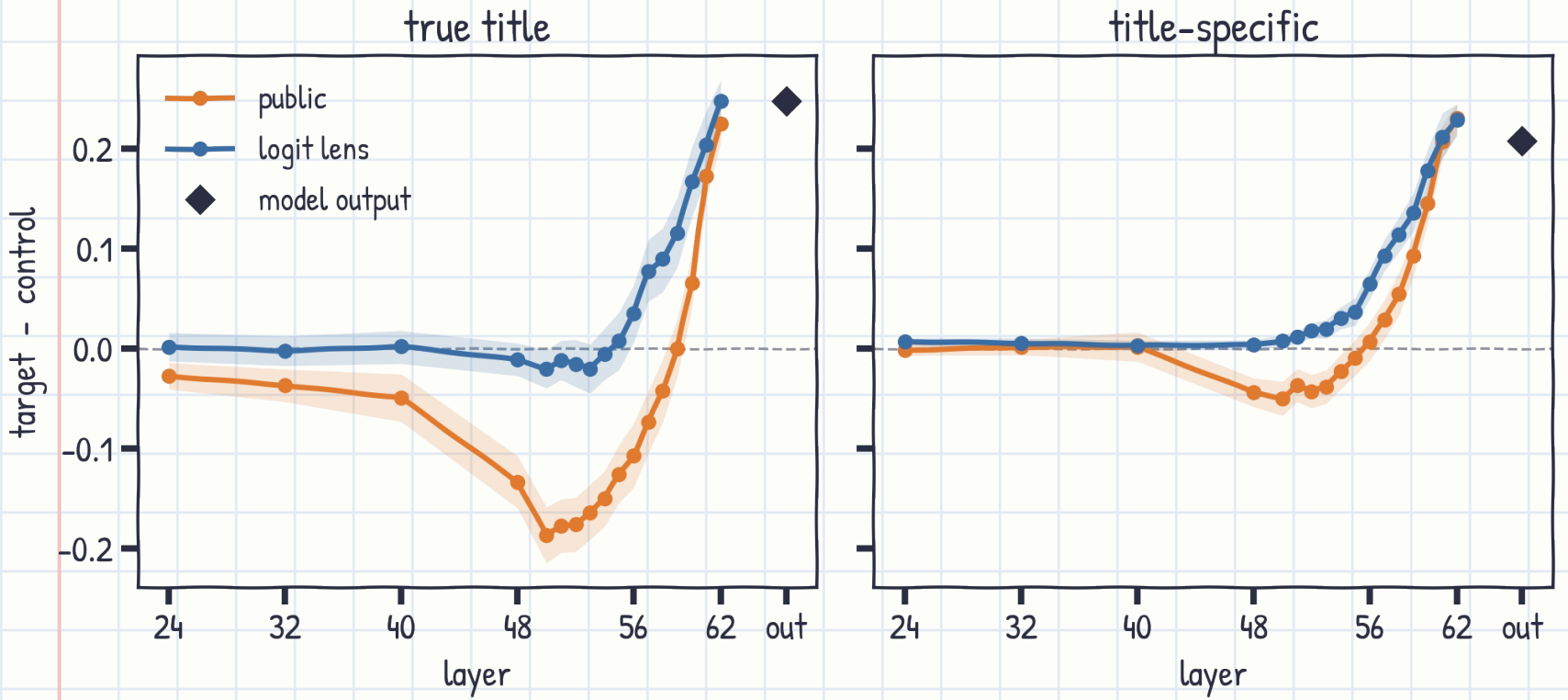
L24–L48: ~0 for both groups

rises at **L56**

L62: target +0.56 vs control +0.24

Target pulls ahead only in the last layers

target − control, within citation bin, 95% CI



Middle layers: **no head start**

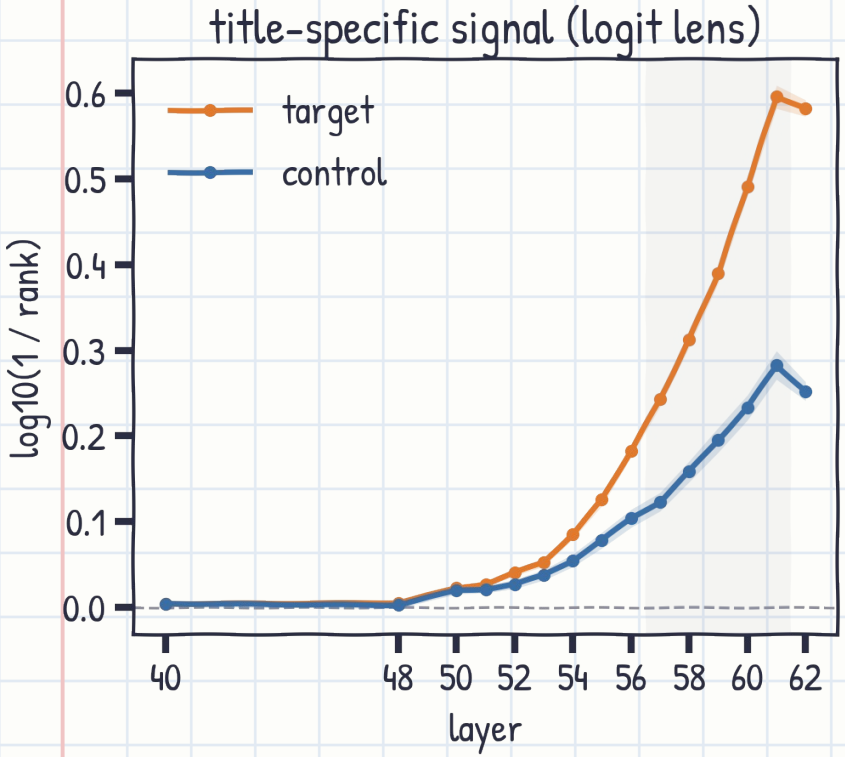
L62 / output:

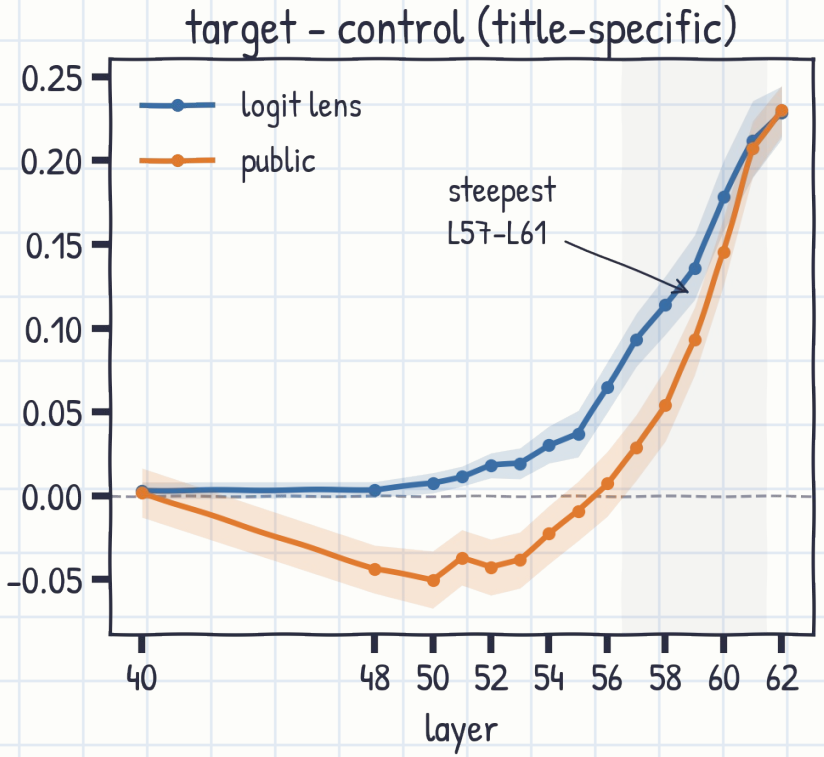
+0.21 to +0.23

almost all title-specific

It builds up over the last dozen layers

fine layers L40–L62 · title-specific signal · 95% CI





Logit lens: target - control

+0.007 at L50

+0.23 at L62

steepest rise **L57–L61**

Stage 11: intervene, do not just read

593 target + 593 control papers (balanced within citation bins) · all 9,108 for probe and chain

patch

copy the true-title hidden state into the false-title run at one layer

knock out

block attention to the title tokens from layer l on

probe

linear map from the last title token to the output state

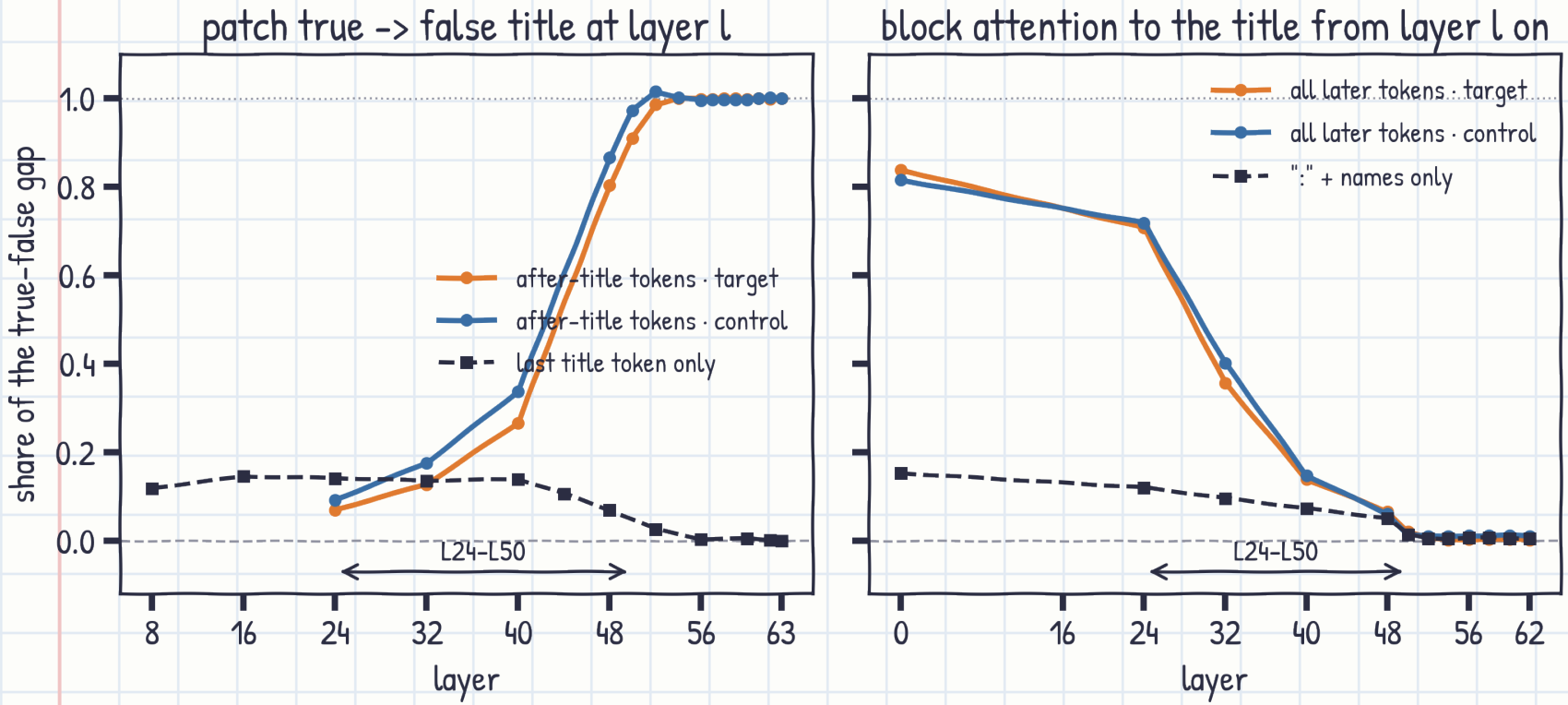
chain

"Authors: A1, A2, ..." instead of each author alone

Score: rank of the correct name token at the model output; "share of the true-false gap" = (intervened − false) / (true − false).

The title is read in L24-L50, early

model output · share of the true-false gap · CIs about +/-0.02



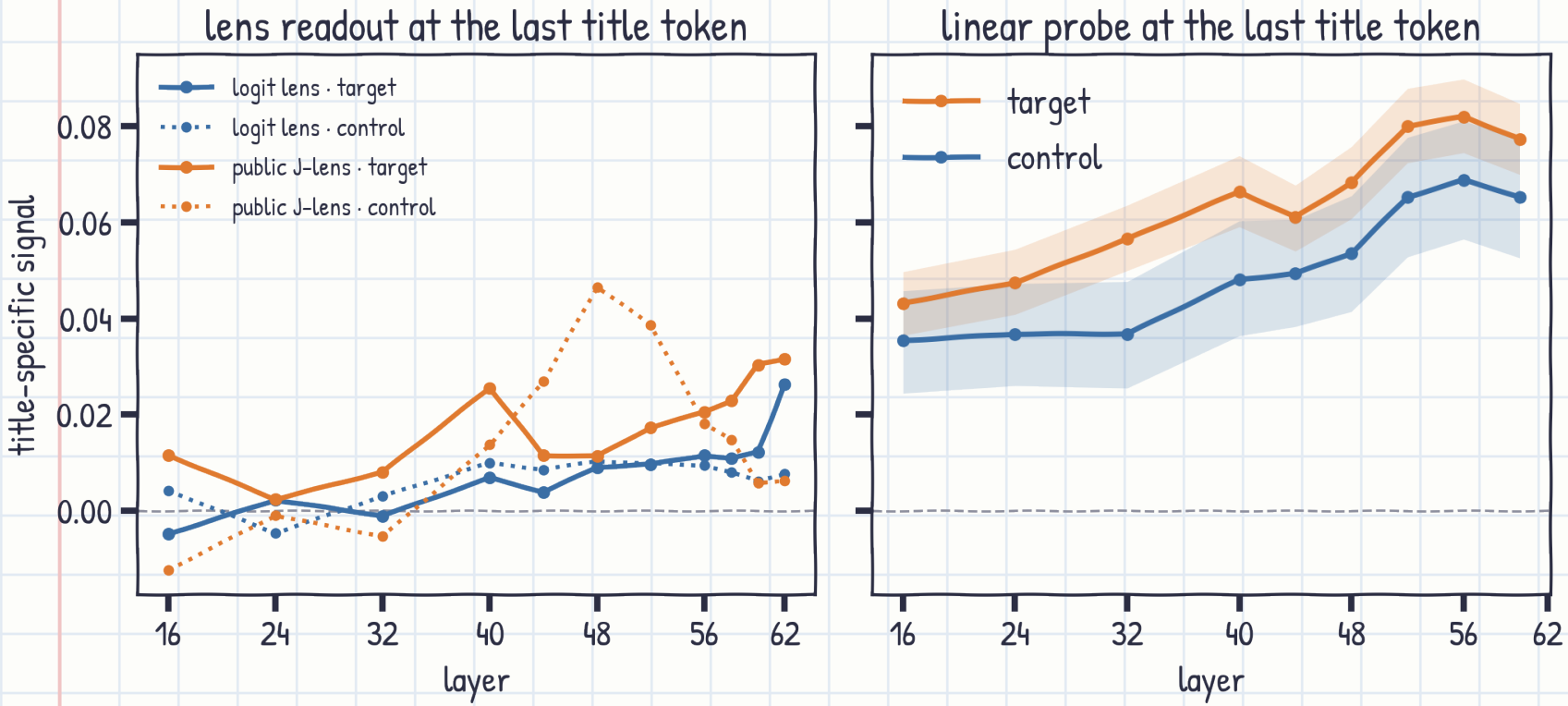
L52+: title no longer needed

info moved to the tokens **right after** the title

same timing for target and control

Lenses cannot read it at the title token

title-specific signal at the last title token · all 9,108 papers



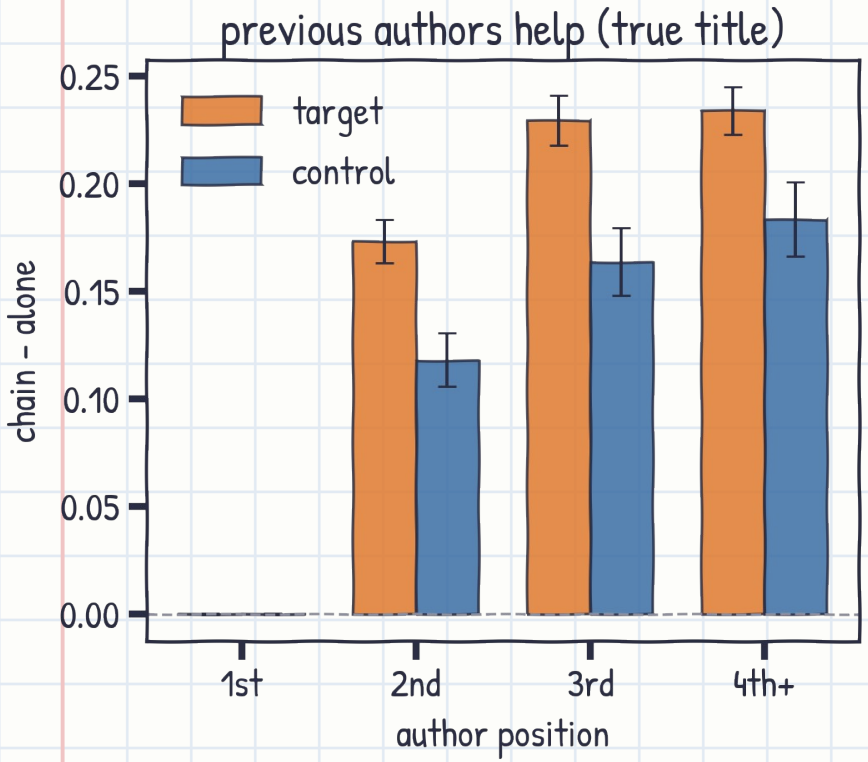
lenses: ~0.01–0.03

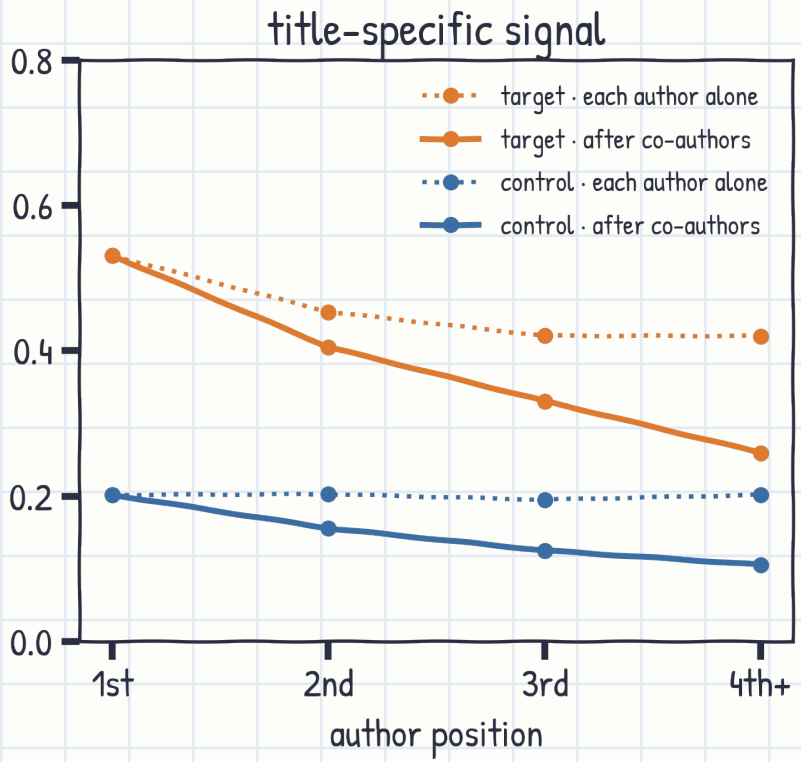
probe: 0.04 -> 0.08

target higher; paper-specific (next slide)

Co-authors help, and partly replace the title

model output · all 9,108 papers · 95% CI





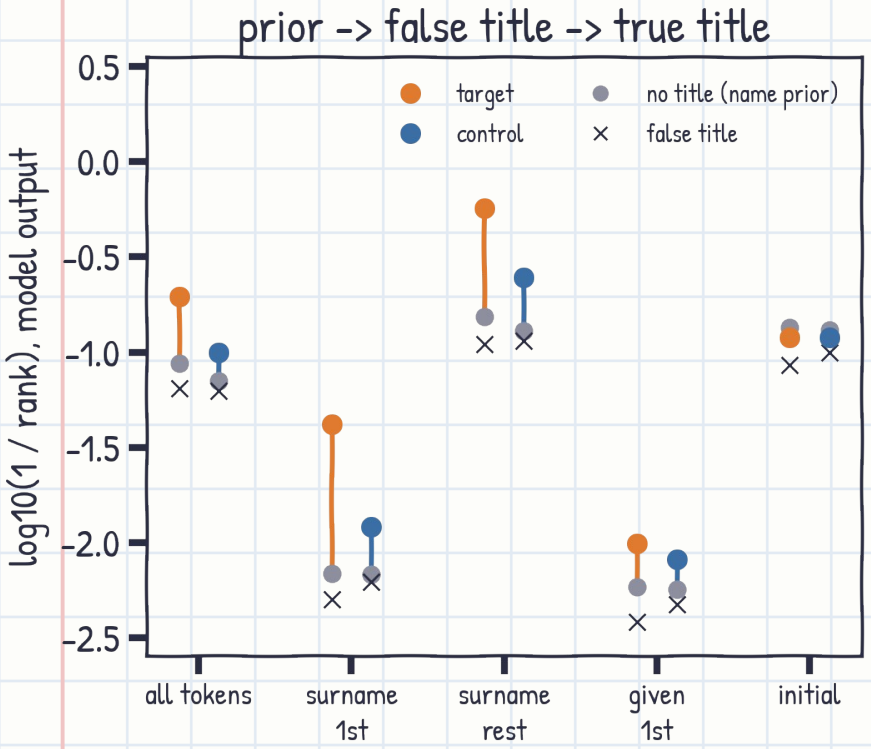
later authors +0.17 to +0.23

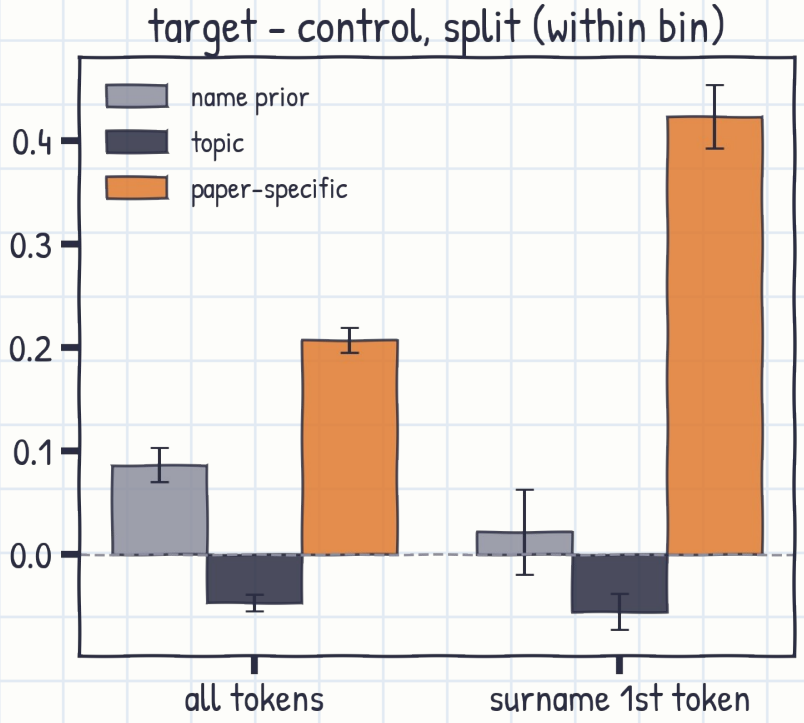
title-specific part **shrinks**

target > control still

How much is just the name prior?

model output · no title vs false title vs true title · all 9,108 papers





no title: rank ~12

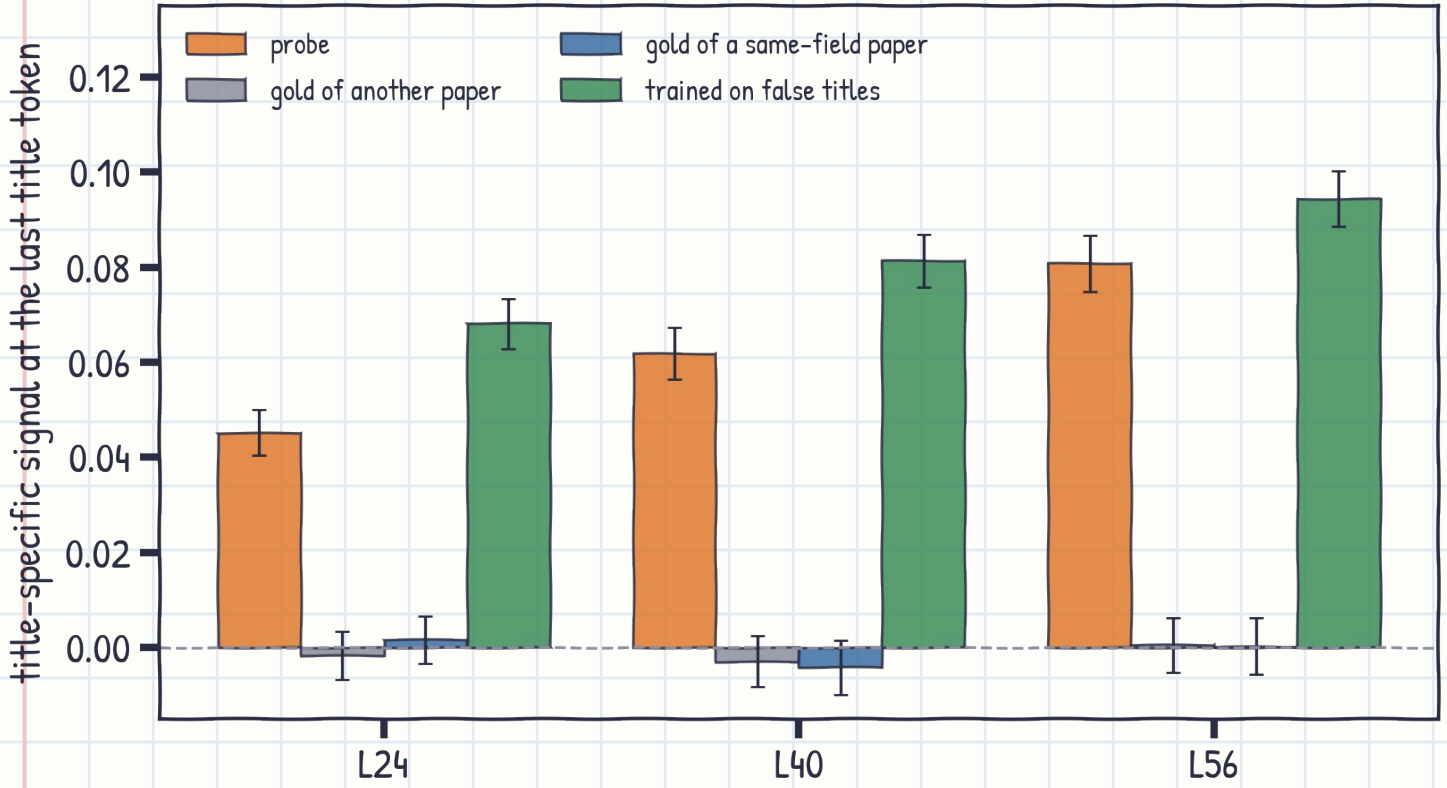
target - control: +0.09 prior, +0.21 paper

surname 1st: prior ~0, paper +0.42

prior vs recognition rho **0.09**

The probe signal is paper-specific

last title token · 5-fold ridge probe · controls · all 9,108 papers



another paper's authors: 0

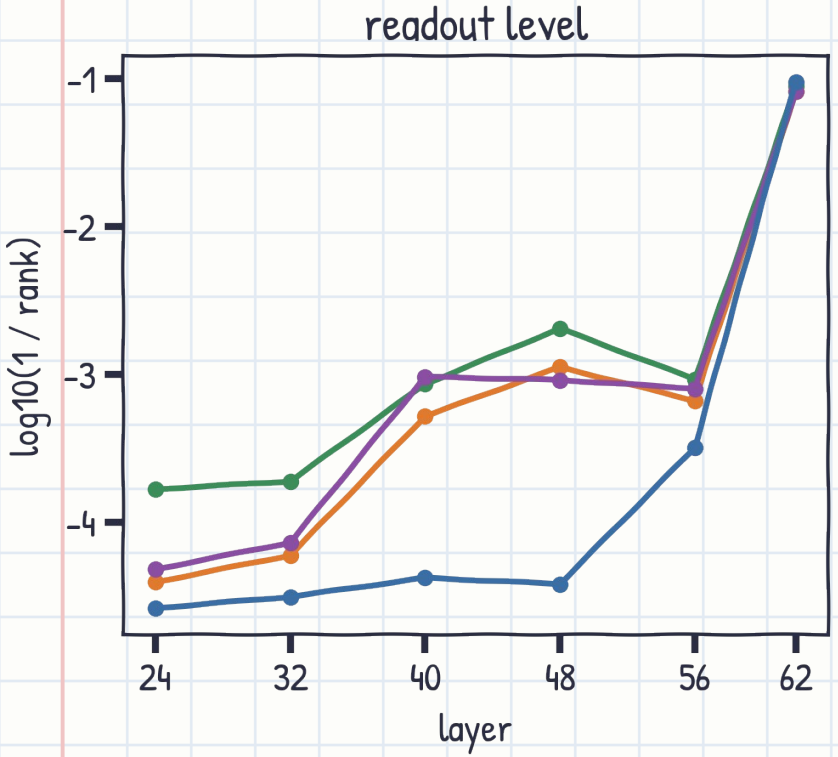
same-field paper: 0

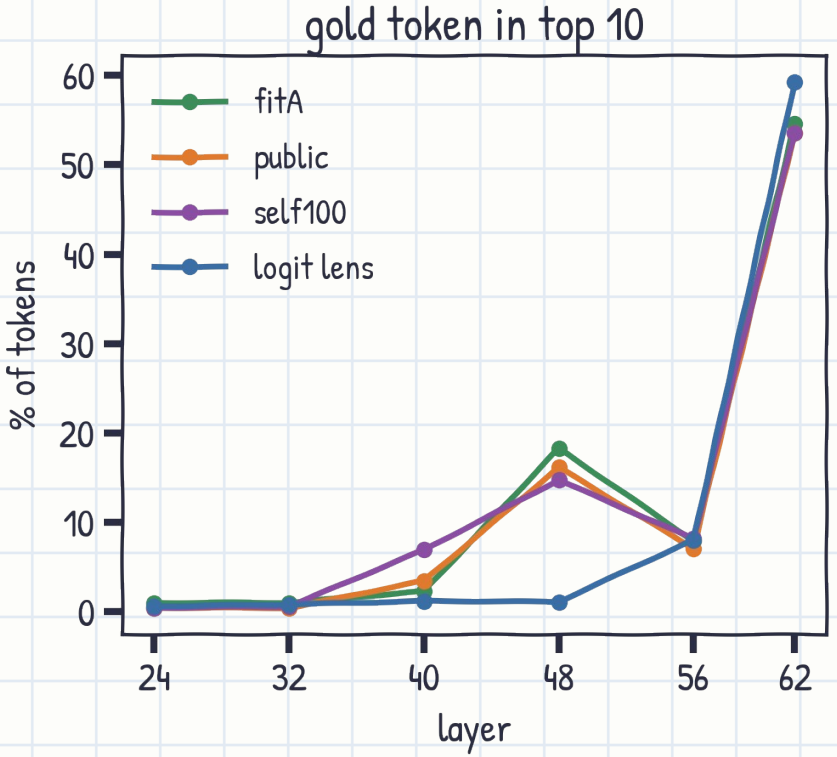
so it is this paper, already at L24

lenses cannot read it

Which lens reads best?

all papers · true title





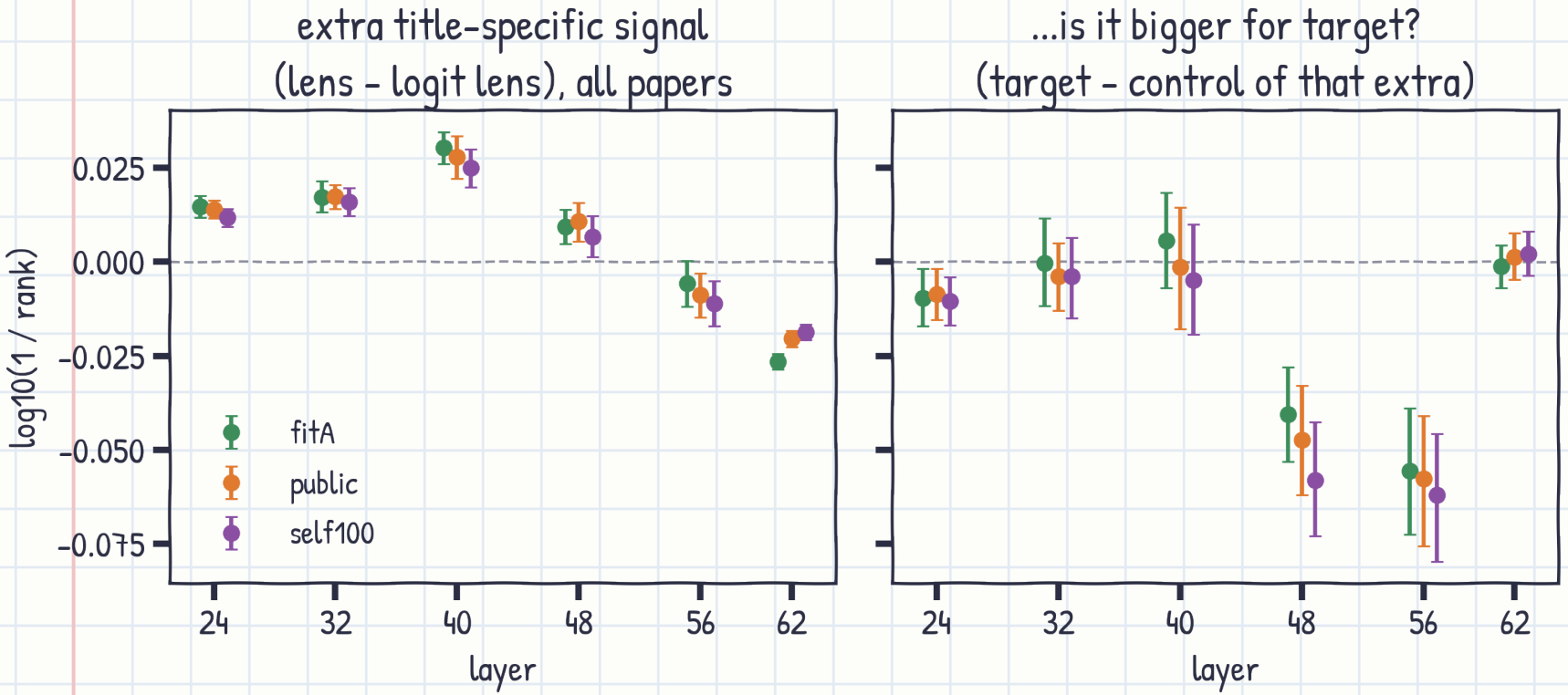
Middle layers:

fitA best; at L48 15–18% top-10 vs 1% for logit

L62: logit lens best (59%)

Do lenses reveal hidden paper knowledge?

paired differences, 95% CI



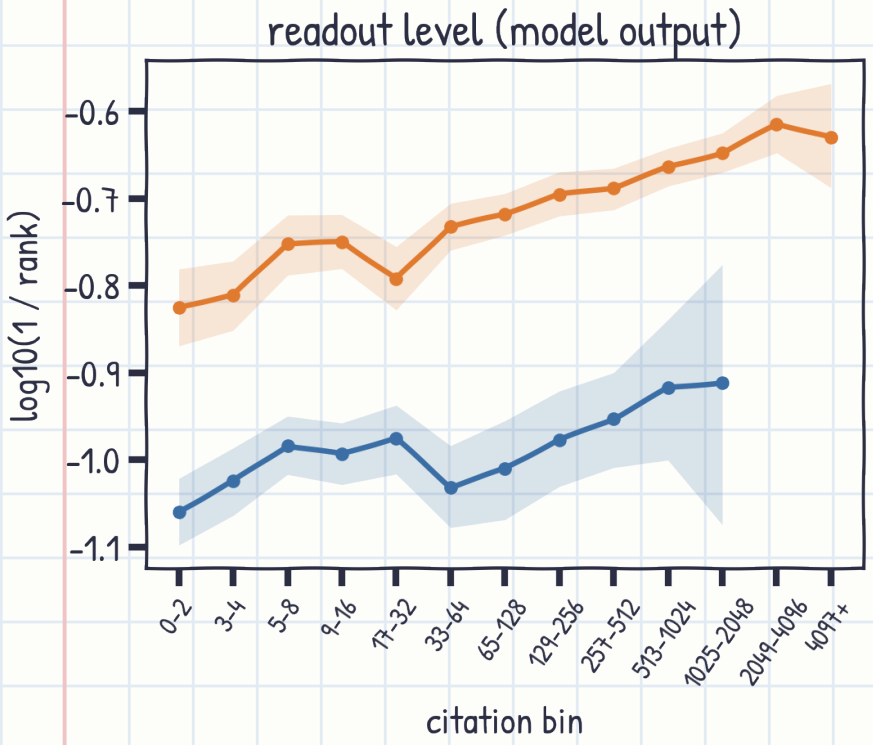
Extra signal: **+0.01 to +0.03**

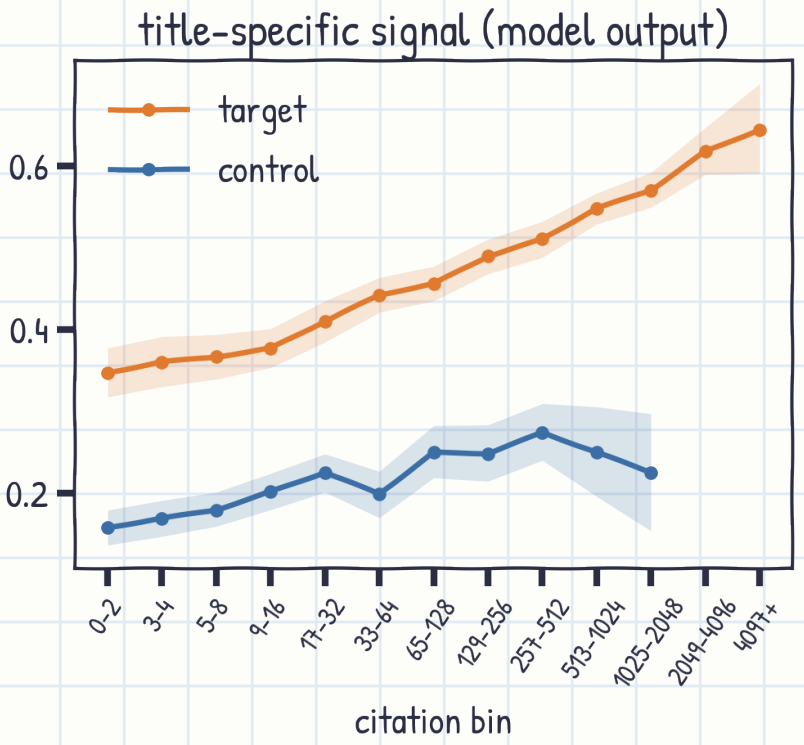
vs ~0.44 at L62

Not bigger for target
(even negative at L48–56)

More citations, more paper knowledge

model output · bins with < 10 papers omitted





Target: +0.35 -> +0.64

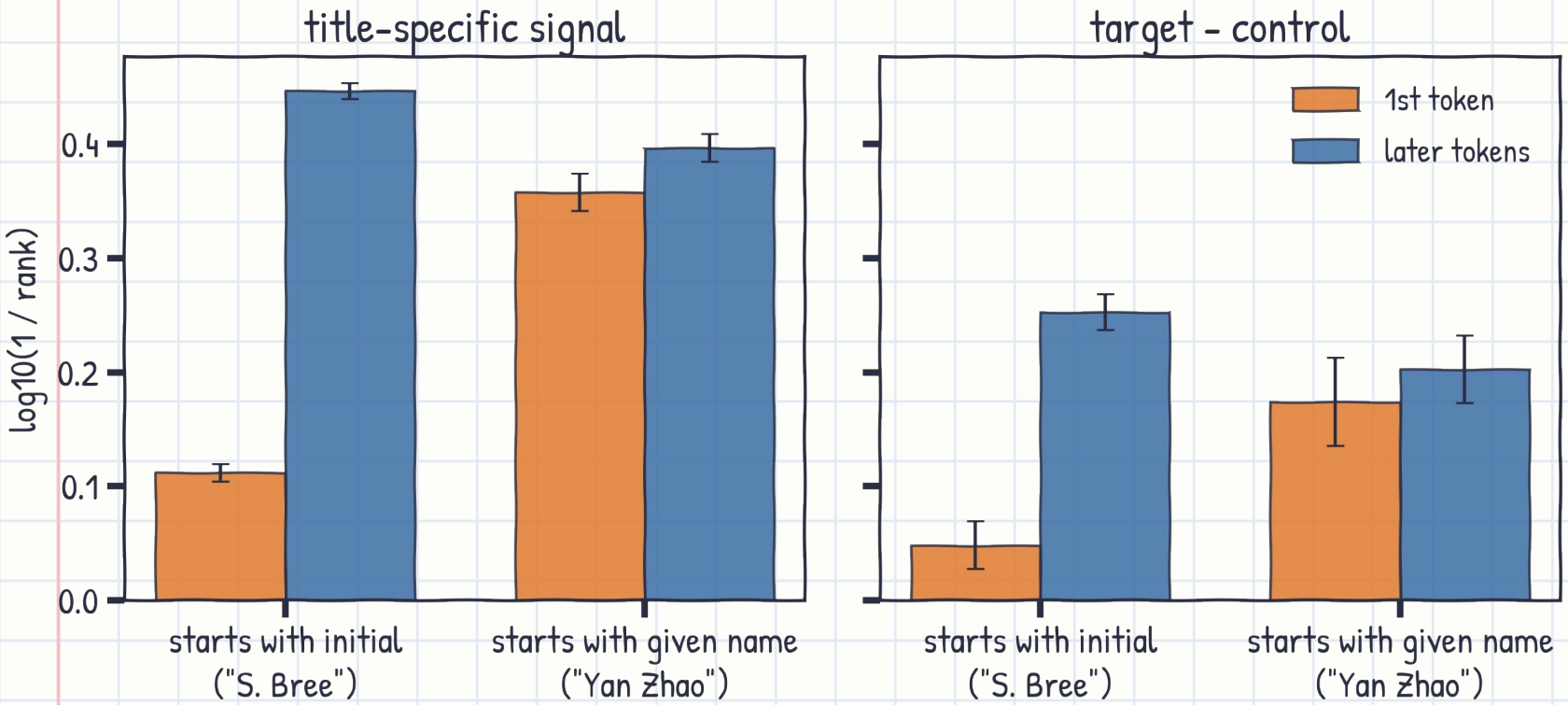
Spearman (title-specific)

0.45 all · 0.33 target · 0.19 control

Target > control in every bin

First token or the rest of the name?

model output · split by name format

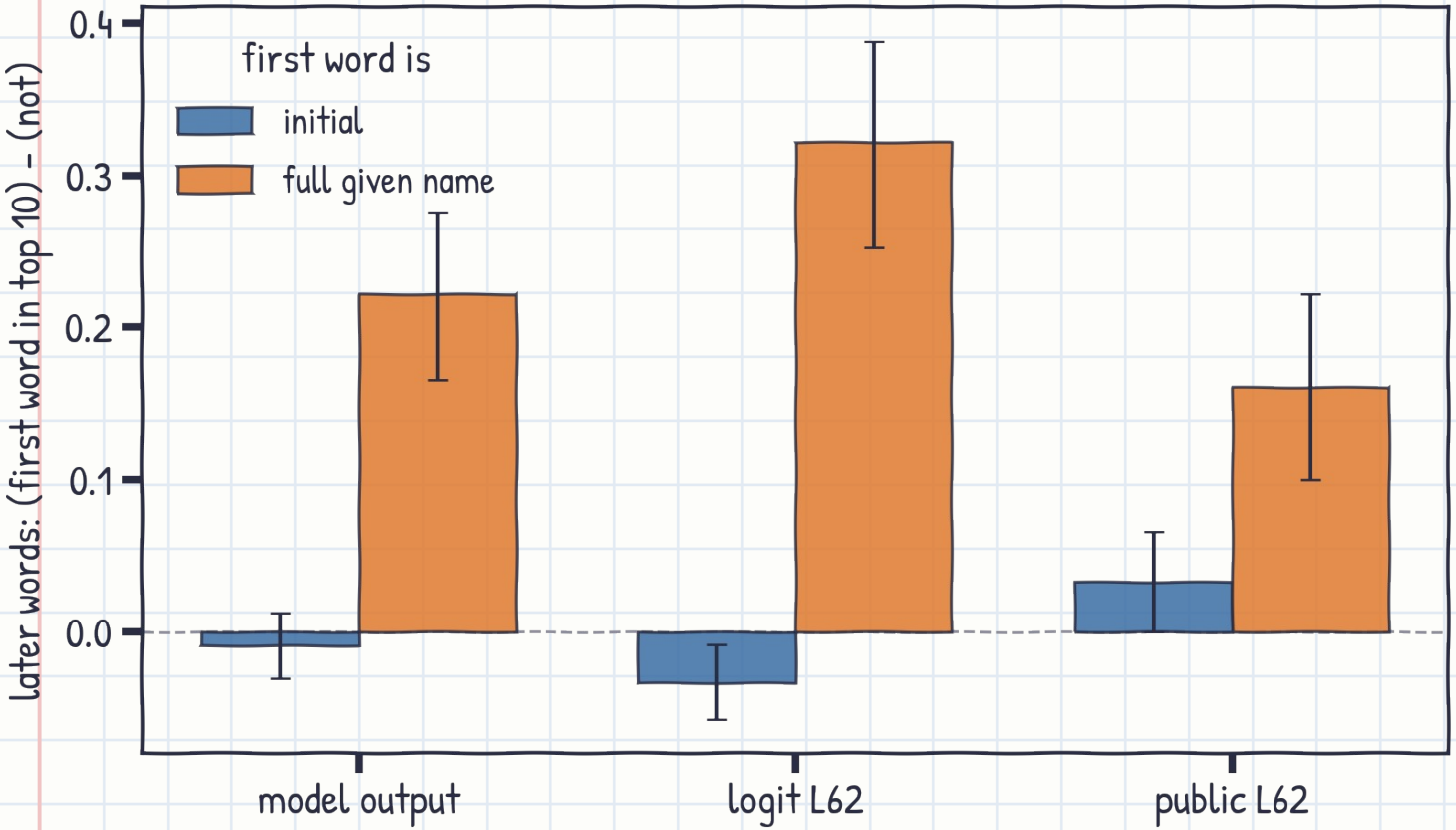


Initial: carries little; signal is on the surname

Full given name: carries about as much as the surname

Knowing the given name helps with the surname

true title · later words when the first word is / is not in the top 10



Given name in top 10

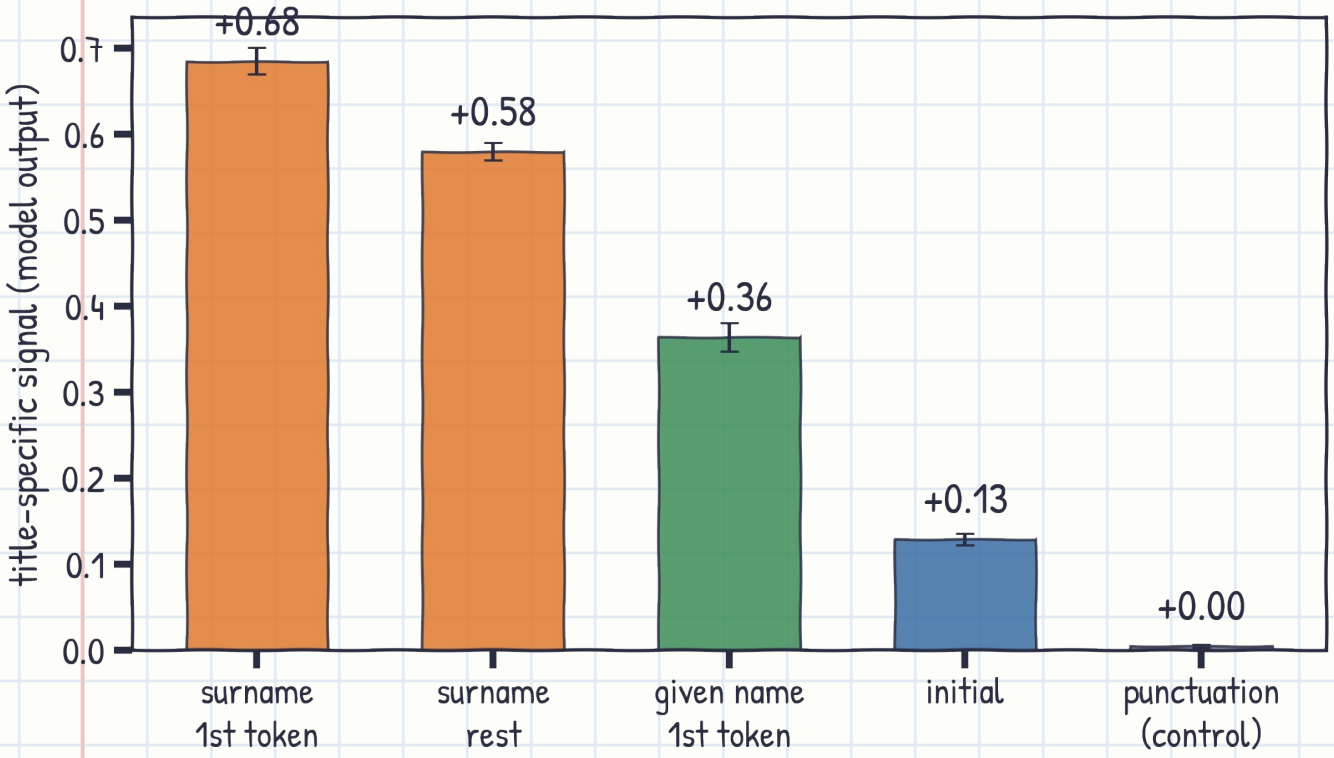
surname +0.22 (output)

+0.32 (logit L62)

Initial in top 10: no help

Where the signal sits, and a dose-response

model output





Right: R0/R1 papers, centred within citation bin; Spearman rho = 0.27, p < 0.001.

What we learned

Late — Paper-specific author knowledge builds up from about L50, fastest over L57–L61, peaking at L61–L62.

Read early — Causally, the title is read in L24–L50 and moved into the tokens after it; after L52 it is no longer needed.

No head start — Target papers are not ahead of control in middle layers; the +0.21 gap is at the end.

Real knowledge — Name prior is ~1/3 of the gap overall but ~0 on surnames; the rest is paper-specific and grows with citations.

Lenses agree — Jacobian lenses read more in middle layers, but nothing target-specific.

Name format — Initials carry little; full given names carry as much as surnames and help predict them.

Caveats and next steps

Caveats

- Readout is correlational; patching/knockout fix the timing, not the mechanism inside L24–L50
- Parity tolerance raised 0.25 -> 0.5 after the fact
- Groups are model-specific; low-citation "recognition" is near chance
- Few control papers above 1,000 citations

Next

1. Patch single positions (quote, year, "Authors:") to find which token carries the paper
2. Patch / probe per layer inside L24–L50 to see when it is computed
3. Repeat on a second model